A Project Report

On

"Learning Management system with smart study planner"

Submitted in partial fulfillment of the requirements
For the award of the degree of Bachelor of Technology in Computer Science & Engineering

AKS UNIVERSITY, SATNA

B.Tech (CSE) 7th Semester

Submitted by

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Under The Guidance of

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CERTIFICATE

This certify that the project report entitled "**Learning Management system with smart study planner**" submitted by partial fulfillment of the requirement for the degree of Bachelor of Technology in Computer Science & Engineering in **April-May 2026** AKS University, Satna is a bonafide project work carried out by **Mo. Suhail, B2255R10106185,** under my supervision. The supervisor has approved the subject of the project report. This is also to certify that it is his/her original work and no part of this project is report has been submitted for any other degree/diploma.

All the assistance and help received during the course of the investigation has been duly acknowledged.

1. I am satisfied that the report presented by **Mo. Suhail** is worthy of consideration for award of the degree.
2. I certify:
   i) That he/she pursued the prescribed course of research. ii) That he/she

   bears good moral character.

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Date:…/…./………

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Prof. (Dr.) Akhilesh A. Waoo **Dr. Chandra Shekhar Gautam** Dean Computer Science and Engineering, (Head CSE, AKS University) AKS University

SELF DECLARATION

I hereby declare that the work presented in this project entitled **"*Learning Management system with smart study planner* "** towards the partial fulfilment of the requirement for the award of **Degree in B.Tech** in Department of Computer Science, **AKS University, Satna (M.P.)** is an authentic record of my own work.

I have not submitted the matter embodied in the project for the award of any other degree or diploma to any other institute or university.

……………………………..

Signature of Candidate

Mo. Suhail

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It is a great opportunity for me in taking this opportunity to express my sincere thanks and appreciation to my supervisor and Head of Department **Dr. Chandra Shekhar Gautam**. I consider myself lucky enough to have such a great project.FGHGHJ

At this moment of accomplishment, first of all, I pay homage to my guide, Dr. Akhilesh A. Waoo, HoD, CSE - Dr. Chandrashekhar Gautam from AKS University Satna (M.P.). This work would not have been possible without his guidance, support, and encouragement. Under his guidance, I successfully overcame many difficulties and learned a lot.

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ABSTRACT

The Learning Management System with Smart Study Planner (LearnFlow LMS) is a web-based application developed to enhance and streamline the modern learning experience. The primary objective of this system is to provide a centralized, efficient, and user-friendly platform for managing courses, tasks, and study schedules. It enables users to plan their academic activities, access learning materials, and track their performance in an organized manner.

The system offers secure user registration and login functionality using JWT-based authentication and encrypted password storage. It supports role-based access where users can interact with different features such as course management, task handling, and study planning. Key features of the system include an interactive dashboard, smart study planner, course progress tracking, analytics visualization, and task management. These features help users stay organized, improve time management, and enhance productivity.

The project is developed using modern technologies such as React.js for building a responsive and dynamic user interface, FastAPI for backend API development, and MongoDB for efficient data storage and retrieval. Additional tools such as Pydantic for data validation, Motor for asynchronous database operations, and Recharts for data visualization are used to improve performance and usability. The system ensures secure data handling through proper authentication, authorization, and validation mechanisms.

This project reduces manual effort, improves accessibility, and provides a structured approach to learning and planning. It demonstrates the practical implementation of full-stack development concepts in solving real-world academic challenges. The system is scalable, reliable, and suitable for students and educational institutions seeking an efficient digital learning solution.

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Introduction

In the modern era of digital transformation, education has evolved significantly with the integration of technology into learning processes. Traditional classroom-based learning is no longer sufficient to meet the growing demands of students who require flexibility, accessibility, and personalized learning experiences. As a result, Learning Management Systems (LMS) have emerged as powerful tools that bridge the gap between educators and learners by providing a centralized digital platform for managing educational content and activities. The **Learning Management System with Smart Study Planner** is designed as an advanced solution that not only delivers educational resources but also enhances the way students plan and manage their studies.

A Learning Management System is a software application that enables the administration, documentation, tracking, reporting, and delivery of educational courses or training programs. It provides a structured environment where instructors can upload course materials such as video lectures, notes, assignments, and quizzes, while students can access these resources anytime and anywhere. This flexibility is especially beneficial in today's fast-paced world, where students often struggle to balance academics with other responsibilities. By digitizing the learning process, LMS platforms reduce dependency on physical classrooms and allow continuous learning beyond geographical boundaries.

However, despite the availability of learning resources, many students face challenges in managing their time effectively. They often find it difficult to decide what to study, when to study, and how much time to allocate to each subject. This lack of planning can lead to procrastination, missed deadlines, and reduced academic performance. To address this issue, the concept of a **Smart Study Planner** has been integrated into the system. The smart planner acts as an intelligent assistant that helps students organize their daily study schedules, prioritize tasks, and maintain consistency in their learning journey.

The Smart Study Planner is a key feature that differentiates this system from traditional LMS platforms. It analyzes user inputs such as subjects, deadlines, and available study time to generate an optimized study plan. This ensures that students can focus on important topics, complete assignments on time, and prepare effectively for exams. By providing reminders and structured schedules, the planner encourages disciplined study habits and reduces the chances of last-minute stress. This combination of learning management and smart planning creates a more efficient and user-friendly educational environment.

Furthermore, the system is designed to support multiple types of users, including administrators, instructors, and students. Each user has a specific role and set of functionalities within the platform. Administrators manage the overall system, instructors are responsible for creating and managing course content, and students interact with the system to access learning materials and track their progress. This role-based access ensures proper organization and smooth functioning of the platform.

Another important aspect of this project is the ability to track and monitor student performance. The system maintains records of completed tasks, submitted assignments, and overall progress. This allows both students and instructors to evaluate performance and identify areas for improvement. Feedback mechanisms can also be integrated to provide guidance and support, helping students enhance their understanding of subjects.

In addition to improving academic performance, the system promotes self-learning and independence among students. By providing easy access to resources and a structured study plan, learners are encouraged to take responsibility for their own education. This aligns with modern educational approaches that emphasize student-centered learning and continuous skill development.

The development of this project also reflects the growing importance of web-based applications in education. With the widespread availability of the internet and digital devices, students can access the system from various platforms, making learning more convenient and inclusive. The use of modern web technologies ensures that the system is responsive, user-friendly, and scalable to accommodate a large number of users.

Moreover, the integration of planning and learning in a single platform reduces the need for multiple applications. Students no longer have to rely on separate tools for note-taking, scheduling, and task management. Instead, they can perform all these activities within one unified system, saving time and effort. This seamless experience enhances user satisfaction and increases productivity.

Security and data management are also important considerations in this system. User data, including personal information and academic records, must be handled securely to ensure privacy and reliability. Proper authentication and authorization mechanisms are implemented to protect the system from unauthorized access. Additionally, the system is designed to maintain data integrity and ensure smooth performance under different conditions.

In conclusion, the **Learning Management System with Smart Study Planner** is a comprehensive solution that addresses both the learning and planning needs of students. It combines the core functionalities of an LMS with the intelligence of a study planner to create a balanced and effective learning environment. By improving accessibility, organization, and time management, the system aims to enhance the overall educational experience and help students achieve their academic goals efficiently. This project highlights the potential of technology in transforming education and provides a practical approach to modern learning challenges.

Objective of the Project

The primary objective of the **Learning Management System with Smart Study Planner** is to design and develop an integrated digital platform that enhances the quality of education by combining learning resources with intelligent planning tools. In today's competitive and fast-moving academic environment, students often face difficulties in managing their time, organizing study materials, and keeping track of assignments. This project aims to address these challenges by providing a unified system that supports both learning and effective study management.

One of the fundamental objectives of this project is to create a **centralized and accessible learning environment**. The system enables instructors to upload course materials such as video lectures, notes, quizzes, and assignments in a structured format. Students can access these resources anytime and from any location, ensuring flexibility and convenience in learning. This eliminates the dependency on physical classrooms and allows continuous access to educational content, which is especially useful for revision and self-paced learning.

Another major objective is to **improve time management and planning skills among students**. Many learners struggle with organizing their daily study routines, which often leads to missed deadlines and last-minute preparation. The Smart Study Planner integrated into the system is designed to overcome this issue by helping students create personalized study schedules. It considers factors such as subjects, deadlines, workload, and available time to generate an effective plan. This encourages students to follow a disciplined routine and reduces stress during exams or assignment submissions.

The project also aims to **enhance student productivity and academic performance**. By providing a clear structure for learning and planning, the system helps students focus on important tasks and avoid unnecessary distractions. Progress tracking features allow students to monitor completed and pending tasks, giving them a clear understanding of their performance. This self-assessment capability motivates students to improve continuously and achieve better academic results.

Another important objective is to **promote self-learning and independent study habits**. In modern education, it is essential for students to take responsibility for their own learning rather than relying entirely on instructors. The proposed system supports this approach by offering easy access to study materials and tools that help students plan their learning journey. This not only improves their knowledge but also develops important skills such as decision-making, problem-solving, and time management.

The system also focuses on **reducing manual effort and increasing efficiency** in managing educational processes. Traditional methods of handling assignments, attendance, and study schedules are often time-consuming and prone to errors. By automating these processes, the system ensures accuracy, saves time, and simplifies the overall management of academic activities. Instructors can easily upload and update content, while students can submit assignments and receive feedback in an organized manner.

Another key objective is to implement a **role-based access system** that ensures proper organization and security. Different users such as administrators, instructors, and students have specific roles and permissions within the system. Administrators are responsible for managing users and maintaining the platform, instructors handle course content and assignments, and students interact with the system for learning and task completion. This structured approach improves system reliability and prevents unauthorized access.

The project also aims to **provide an interactive and user-friendly interface**. A well-designed interface ensures that users can easily navigate the system without confusion. The goal is to make the platform simple, intuitive, and accessible to users with basic technical knowledge. This increases user engagement and encourages regular usage of the system for learning and planning purposes.

Scalability and flexibility are also important objectives of this project. The system is designed in such a way that it can handle a growing number of users and can be extended with additional features in the future. For example, new functionalities such as notifications, performance analytics, or integration with other educational tools can be added as needed. This ensures that the system remains useful and adaptable over time.

Another objective is to **improve communication between instructors and students**. The platform can include features such as announcements, feedback, and updates that keep students informed about their courses and tasks. Effective communication helps in reducing confusion, improving clarity, and creating a better learning environment.

The project also emphasizes **data management and security**. Protecting user data is a critical aspect of any digital system. The platform ensures secure storage of information such as user details, academic records, and submitted assignments. Proper authentication and authorization mechanisms are implemented to prevent unauthorized access and maintain data privacy.

Furthermore, the system aims to **integrate multiple functionalities into a single platform**. Instead of using separate tools for learning, scheduling, and task management, students can perform all these activities within one system. This not only saves time but also provides a seamless and organized experience, making it easier for users to manage their academic responsibilities.

In conclusion, the objective of the **Learning Management System with Smart Study Planner** is to develop a comprehensive solution that improves the efficiency, accessibility, and effectiveness of the learning process. By combining educational content with intelligent planning tools, the system helps students manage their time, track their progress, and achieve their academic goals. It also supports instructors in delivering content more effectively and simplifies the overall management of educational activities. This project reflects the importance of technology in modern education and aims to create a smart, organized, and productive learning environment for all users.

Scope of the Projects

The scope of the **Learning Management System with Smart Study Planner** outlines the boundaries, functionalities, and overall capabilities of the proposed system. This project is designed to create a comprehensive web-based platform that integrates digital learning with intelligent study planning. It focuses on simplifying the academic process for students and instructors by providing a centralized, structured, and efficient system for managing educational activities.

The primary scope of this project includes the development of a **centralized learning platform** where all academic resources and activities are managed in one place. Instructors can upload course materials such as video lectures, notes, assignments, and quizzes, while students can access these materials at their convenience. This eliminates the need for multiple platforms and ensures that all learning-related content is organized and easily accessible. The system supports continuous learning by allowing users to revisit content whenever required, making it highly beneficial for revision and self-paced study.

Another important area within the scope is the **Smart Study Planner**, which plays a crucial role in improving students' time management skills. The planner allows students to create customized study schedules based on their subjects, deadlines, and available time. It helps in prioritizing tasks and ensures that important topics are covered on time. By providing reminders and structured plans, the system encourages consistency and discipline in study habits. This feature is especially useful for students who struggle with organizing their daily academic activities.

The system also includes **user management and role-based access control**, which ensures that different users have specific permissions and responsibilities. The platform supports three main types of users: administrators, instructors, and students. Administrators are responsible for managing the system, including user accounts and overall functionality. Instructors can create and manage course content, assign tasks, and monitor student progress. Students can access learning materials, complete assignments, and track their performance. This structured approach ensures smooth operation and enhances system security.

Within the scope of the project, **assignment and task management** is another key feature. Instructors can assign tasks and set deadlines, while students can submit their work through the system. The platform keeps track of submissions and completion status, making it easier to manage academic activities. This reduces manual effort and improves efficiency in handling assignments. It also allows instructors to provide feedback, which helps students improve their understanding and performance.

The project also covers **progress tracking and performance monitoring**, which is essential for evaluating student learning. The system maintains records of completed lessons, submitted assignments, and overall progress. Students can view their performance and identify areas where they need improvement. Instructors can use this data to monitor student engagement and provide guidance when necessary. This feature promotes continuous improvement and helps maintain academic standards.

Another important aspect of the scope is **accessibility and flexibility**. The system is designed as a web-based application, which means it can be accessed through internet-enabled devices such as computers, laptops, and smartphones. This ensures that users can access the platform anytime and from anywhere. It supports remote learning and provides flexibility to students who may not always be able to attend physical classes. This makes the system suitable for a wide range of users, including students from different locations and backgrounds.

The scope also includes **basic communication features** that enhance interaction between instructors and students. The platform can provide announcements, notifications, and updates regarding assignments, deadlines, and course activities. This ensures that students stay informed and do not miss important information. Effective communication plays a vital role in maintaining a smooth learning process and improving coordination between users.

In addition, the system focuses on **data management and security**. All user data, including personal information and academic records, is stored securely within the system. Authentication and authorization mechanisms are implemented to ensure that only authorized users can access specific data. This helps in maintaining privacy and protecting sensitive information from unauthorized access.

The project also considers **system efficiency and usability**. The interface is designed to be simple, user-friendly, and easy to navigate. Users with basic technical knowledge can operate the system without difficulty. This improves user experience and encourages regular usage of the platform. The system aims to minimize complexity while providing all essential functionalities in an organized manner.

Another key part of the scope is **integration of multiple functionalities into a single platform**. Instead of using separate tools for learning, scheduling, and task management, the system combines all these features into one application. This reduces confusion, saves time, and improves productivity. Students can manage their entire academic workflow within a single system, making it more efficient and convenient.

The scope of the project also includes **future scalability and enhancement possibilities**. While the current system focuses on core functionalities, it is designed in such a way that additional features can be added in the future. For example, advanced analytics, AI-based study recommendations, mobile application support, and integration with other educational platforms can be implemented later. This ensures that the system remains adaptable and relevant to future needs.

However, it is also important to define the **limitations of the project**. The initial version of the system may not include advanced features such as live video conferencing, real-time collaboration tools, or highly sophisticated artificial intelligence capabilities. These features require additional resources and can be considered for future development. The current scope focuses on delivering a stable, functional, and efficient system with essential features.

Furthermore, the system is intended for **academic and educational purposes only**. It is not designed to handle non-educational activities or commercial operations. The focus remains on improving learning outcomes and supporting students in their academic journey.

In conclusion, the scope of the **Learning Management System with Smart Study Planner** is broad and covers all major aspects of digital learning and study management. It includes features such as content management, task handling, scheduling, progress tracking, and communication, all within a single platform. The system aims to provide a structured, secure, and user-friendly environment that enhances the learning experience. Although the current scope focuses on core functionalities, the system has strong potential for future expansion and improvement, making it a valuable solution for modern education.

Problem Definition

A library is an essential academic resource for students, teachers, and researchers. It provides access to books, journals, and learning materials that support education and knowledge development. However, in many institutions, library operations are still carried out manually using handwritten registers, physical logbooks, and paper-based record systems. While this method has been used for decades, it faces several serious challenges, especially with increasing numbers of books and users. These limitations create the need for a more advanced, automated, and reliable library management system. The **Definition of the Problem** explains the issues associated with traditional manual library systems and why a digital alternative is necessary.

The biggest problem with the manual system is **inefficiency**. Every process—whether it is issuing books, returning them, maintaining user records, or searching for information—requires significant time and effort. Librarians must manually write book details, verify user credentials, and update registers, which slows down the overall workflow. During peak hours, such as before exams, long queues form, causing frustration among students and staff members.

Another major issue is **inaccuracy**. Human errors are common when data is written by hand. Mistakes such as incorrect book titles, mismatched issue dates, unreadable handwriting, skipped entries, and duplicate records can easily occur. These errors accumulate over time and make it difficult to maintain reliable and accurate information. A single incorrect entry can lead to confusion about book availability or user responsibility.

The manual record-keeping system also struggles with **tracking book availability**. Since all information is stored in registers, librarians and users often have difficulty determining whether a book is available, issued, overdue, or missing. This lack of real-time information leads to delays and inefficiency. Students may search for a book that is already issued, or a librarian may unknowingly issue a book that should be reserved for another user.

Searching for books in a traditional library is also a time-consuming task. Users must manually check shelves or rely on librarians to search through multiple registers. When the collection becomes large, this process becomes slow and frustrating. There is no quick way to search based on keywords, authors, categories, or subjects, which limits accessibility and convenience.

Another significant problem is the **lack of automated reminders and overdue tracking**. In a manual system, it becomes nearly impossible to monitor due dates for hundreds of issued books. As a result, users may forget to return books on time, leading to delayed submissions and potential loss of resources. Librarians cannot manually track every overdue book, which results in irregular penalties, miscommunication, and financial loss to the library.

Additionally, **report generation becomes extremely difficult** in manual systems. Reports such as daily transactions, book issuance logs, overdue lists, user history, or stock availability require extensive manual effort. Librarians often need to go through multiple registers and compile information manually, which is time-consuming and prone to mistakes. This lack of timely reporting makes it difficult for administrators to plan purchases or monitor library usage patterns.

Another major concern is **poor data security**. Physical registers can be damaged, lost, torn, or altered intentionally or accidentally. Fire, water damage, or simple wear and tear over time can destroy years of important records. There is no secure backup, no version control, and no protection against unauthorized access. Sensitive data such as user information or transaction history is vulnerable in a manual setup.

The **absence of centralized data storage** also creates problems. Information is scattered across multiple registers, making it hard to retrieve, update, or organize. Librarians need to search through different books for different types of information, which leads to delays and inefficiency. There is no unified database that combines books, users, and transactions under one system.

As the number of users increases, **workload on librarians also grows significantly**. Manual processes cannot scale efficiently when thousands of students use the library. Librarians struggle to manage book circulation, maintain accurate records, and serve users simultaneously. This increased workload leads to stress, slow service, and operational inefficiencies.

Another issue is **difficulty in monitoring library usage**. Without proper data, it is hard to analyze which books are used most frequently, which subjects are in high demand, and which books are rarely used. This results in poor decision-making regarding resource allocation and purchasing of new books. Libraries may continue buying unnecessary books or fail to invest in high-demand categories due to lack of proper data insights.

Communication gaps are also common in manual systems. Users may not know if a particular book has been returned, issued, or reserved. Librarians may be unable to inform students about due dates, new arrivals, or updated rules. Because there is no direct communication system, users remain unaware of important updates.

Another major problem is the **lack of transparency**. Since all entries are handwritten, users cannot verify their own borrowing history or check real-time status of books. This sometimes leads to disputes between users and librarians, especially during overdue or lost book cases. Without a transparent system, trust and accountability become difficult to maintain.

In addition, **manual systems cannot integrate with modern technologies** like barcode scanning, online reservations, automated notifications, cloud storage, or mobile applications. This limits the growth and modernization of the library. Educational institutions must keep pace with technology to provide a better learning environment, and traditional systems cannot support such enhancements.

Finally, the manual approach fails to meet the expectations of today’s digital generation. Students expect fast, accurate, and easily accessible information. They prefer systems that allow quick search, instant availability status, and smooth operation. The lack of a digital platform reduces user satisfaction and negatively impacts the library’s effectiveness.

In conclusion, the **problem definition** highlights several issues with the traditional manual library system: inefficiency, inaccuracy, slow processes, difficulty in tracking, poor data security, lack of automation, weak reporting, no transparency, and limited scalability. These problems create significant challenges for both librarians and users. Therefore, there is an urgent need for a **Book Library Management System** that can automate operations, secure data, provide instant information, generate useful reports, improve user experience, and support future technological advancements.

Benefit of the Projects

The **Learning Management System with Smart Study Planner** offers a wide range of benefits that improve the overall learning experience for students, instructors, and educational institutions. By combining digital learning with intelligent planning tools, the system creates a structured, efficient, and user-friendly environment that supports academic success and personal development.

One of the most important benefits of this project is **anytime, anywhere access to learning materials**. Students can easily access video lectures, notes, and assignments through the platform without being restricted by location or time. This flexibility allows learners to study at their own pace and revisit topics whenever required, which is especially helpful for better understanding and revision.

Another key benefit is the **Smart Study Planner**, which helps students manage their time effectively. Many students struggle with organizing their study schedules and often delay important tasks. The planner solves this problem by allowing users to create structured schedules based on deadlines and priorities. It ensures that students stay consistent in their studies and complete tasks on time, reducing stress and improving productivity.

The system also enhances **student performance and productivity**. By providing a clear roadmap of tasks and learning materials, students can focus more on important topics and avoid distractions. The progress tracking feature allows them to monitor their achievements and identify areas that need improvement. This continuous evaluation motivates students to perform better academically.

Another major advantage is the **promotion of self-learning and independence**. The platform encourages students to take responsibility for their own learning by giving them full control over their study plans and resources. This helps in developing essential skills such as time management, discipline, and problem-solving, which are important for future careers.

For instructors, the system provides **efficient content management and evaluation tools**. Teachers can easily upload study materials, assign tasks, and track student progress. This reduces manual work and saves time, allowing instructors to focus more on teaching and guiding students. The ability to provide digital feedback also improves communication and helps students understand their mistakes.

The project also helps in **reducing paperwork and manual effort**. Traditional systems rely heavily on physical documents and manual record-keeping, which can be time-consuming and prone to errors. This digital system automates these processes, making them faster, more accurate, and easier to manage.

Another important benefit is **better organization of academic activities**. All information related to courses, assignments, schedules, and progress is stored in a structured manner within the system. This makes it easy for users to access and manage data without confusion, leading to a smoother learning experience.

The system also improves **communication between students and instructors**. Features like notifications and announcements ensure that students are always updated about deadlines, tasks, and important information. This reduces misunderstandings and improves coordination within the learning environment.

In terms of security, the system ensures **safe and secure data management**. User data, including personal details and academic records, is protected through authentication and authorization mechanisms. This ensures privacy and prevents unauthorized access to sensitive information.

Another benefit is the **integration of multiple functionalities into a single platform**. Students do not need to use separate applications for learning, scheduling, and task management. Everything is available in one place, which saves time and improves efficiency.

The system also provides **scalability and flexibility**, allowing it to be enhanced with additional features in the future. For example, advanced analytics, AI-based recommendations, and mobile application support can be added to improve functionality and user experience.

Furthermore, the project helps in **reducing academic stress**. By providing a clear plan and organized structure, students can avoid last-minute preparation and manage their workload effectively. This leads to a more balanced and stress-free learning process.

In conclusion, the **Learning Management System with Smart Study Planner** provides numerous benefits that make learning more accessible, organized, and efficient. It improves time management, enhances productivity, promotes self-learning, and simplifies academic processes. The system not only supports students in achieving their academic goals but also assists instructors in delivering content effectively, making it a valuable solution for modern education.

Main Modules

The **Learning Management System with Smart Study Planner** is designed using a modular approach, where the entire system is divided into multiple functional units called modules. Each module performs a specific task and works together with other modules to ensure smooth and efficient operation of the system. This modular structure improves maintainability, scalability, and clarity in system design. The major modules of the system are described below in detail.

1. Authentication and User Management Module

The Authentication and User Management Module is one of the most important components of the system. It is responsible for handling user registration, login, and access control. This module ensures that only authorized users can access the system and perform specific actions based on their roles.

Users can register themselves by providing necessary details such as name, email, and password. The system securely stores user credentials using encryption techniques. During login, the system verifies user credentials and generates a secure session or token, allowing access to the platform.

This module also implements **role-based access control**, where different users such as administrators, instructors, and students have different permissions. For example, instructors can upload content and create assignments, while students can only view materials and submit tasks. Administrators manage all users and system settings.

Additionally, the module manages user profiles, allowing users to update their personal information. It also ensures data security by preventing unauthorized access and protecting sensitive user data.

2. Course Management Module

The Course Management Module is responsible for managing all course-related activities within the system. It allows instructors to create, update, and organize courses in a structured manner.

Instructors can upload various types of learning materials such as video lectures, PDF notes, and presentations. They can organize content into different sections or topics, making it easier for students to follow the course structure. This helps in maintaining a logical flow of information and improves the learning experience.

Students can browse available courses, enroll in them, and access the provided materials. The module ensures that students can easily navigate through the course content without confusion. It also supports updating and modifying course details whenever required.

This module plays a key role in delivering educational content effectively and ensuring that all learning resources are well-organized and easily accessible.

3. Task and Assignment Management Module

The Task and Assignment Management Module handles the creation, distribution, and submission of assignments. It is essential for maintaining continuous assessment and tracking student performance.

Instructors can create assignments and set deadlines for submission. These assignments are then visible to students through their dashboards. Students can complete the tasks and submit their work through the system.

The module keeps track of all submissions and maintains records of completed and pending tasks. Instructors can review submissions and provide feedback to students, which helps in improving their understanding of the subject.

This module reduces manual work involved in handling assignments and ensures that all academic tasks are managed efficiently and systematically.

4. Smart Study Planner Module

The Smart Study Planner Module is a unique and advanced feature of the system that helps students organize their study schedules effectively. It acts as an intelligent assistant that guides students in planning their daily, weekly, or monthly study routines.

Students can input their subjects, deadlines, and available study time. Based on this information, the system generates a structured study plan that prioritizes tasks according to their importance and urgency. This helps students focus on important topics and manage their time efficiently.

The planner also provides reminders and notifications to ensure that students follow their schedules consistently. It helps in reducing procrastination and encourages disciplined study habits.

This module is highly beneficial for students who struggle with time management and organization. By providing a clear plan, it reduces stress and improves overall productivity.

5. Dashboard and Analytics Module

The Dashboard and Analytics Module provides a visual representation of user data and system activities. It serves as the main interface where users can view important information such as tasks, progress, and performance metrics.

The dashboard displays an overview of completed and pending tasks, upcoming deadlines, and overall progress. It helps students quickly understand their current status and plan their activities accordingly.

The analytics feature provides insights into student performance, such as task completion rates and learning trends. This information can be used by both students and instructors to identify strengths and areas for improvement.

This module enhances decision-making by providing clear and meaningful data in an easy-to-understand format.

6. Communication and Notification Module

The Communication and Notification Module ensures smooth interaction between students and instructors. It provides features such as announcements, alerts, and reminders.

Instructors can send notifications regarding assignments, deadlines, and course updates. Students receive these notifications and stay informed about important activities. This reduces the chances of missing deadlines and improves coordination.

The module may also include messaging or feedback features that allow users to communicate directly. This improves engagement and creates a more interactive learning environment.

7. Database Management Module

The Database Management Module is responsible for storing and managing all system data. It handles user information, course details, assignments, planner data, and performance records.

The system uses a database to store data in an organized and structured manner. It ensures data consistency, reliability, and quick retrieval when needed. Proper indexing and optimization techniques are used to improve performance.

Security is also an important aspect of this module. It ensures that sensitive data is protected and only accessible to authorized users. Regular backups and data validation techniques help maintain data integrity.

8. API and Backend Services Module

The API and Backend Services Module acts as the bridge between the frontend and the database. It processes user requests, performs business logic, and returns appropriate responses.

When a user performs an action, such as logging in or fetching tasks, the request is sent to the backend. The backend processes the request, interacts with the database, and sends the required data back to the frontend.

This module ensures smooth communication between different parts of the system and plays a crucial role in overall functionality and performance.

9. Frontend User Interface Module

The Frontend User Interface Module is responsible for the visual appearance and interaction of the system. It provides a user-friendly interface that allows users to easily navigate through the platform.

The interface includes pages such as dashboard, planner, courses, analytics, and login screens. It is designed to be responsive, ensuring compatibility with different devices such as desktops, tablets, and smartphones.

A well-designed interface improves user experience and encourages regular usage of the system.

Technical Overview

1. Technology Stack (Full Stack)

The system uses a modern full-stack architecture which includes MongoDB, FastAPI, React.js, and Python.

• **MongoDB** is used as a NoSQL database to store users, tasks, courses, and planner data securely.

• **FastAPI** is used to create backend APIs, routing, and server-side logic with high performance.

• **React.js** is used to build the user interface with reusable and dynamic components.

• **Python** is used as a backend programming language for executing server-side logic.

• **Uvicorn** is used as a high-performance server to run the FastAPI application efficiently.

2. Additional Backend Technologies

The backend includes several supporting technologies and libraries.

• **Motor** is used as an asynchronous MongoDB driver for fast and efficient database operations.

• **Pydantic** is used for data validation and defining request and response schemas.

• **JWT (JSON Web Token)** is used for secure authentication and authorization.

• **Passlib (bcrypt)** is used for password hashing and secure authentication.

• **CORS** is used to allow safe communication between frontend and backend.

• **dotenv (.env)** is used for storing sensitive configuration variables securely.

• **Python-JOSE** is used for handling JWT token encoding and decoding.

3. Frontend Technologies

The frontend is designed using modern and responsive tools.

- **HTML** is used for creating the structure of web pages.
- **React.js** is used for building dynamic and fast-loading UI components.
- **Vite** is used as a fast development and build tool.
- **Tailwind CSS** is used for modern, responsive, and utility-based styling.
- **React Router** is used for navigation between different pages.
- **Recharts** is used for displaying analytics and performance data in graphical form.
- **Lucide React** is used for adding modern and customizable icons to the user interface.

System Configuration

1. Hardware Requirements

a) Minimum Hardware Requirements

- Processor: Intel Core i3 or equivalent.
- RAM: 4 GB.
- Hard Disk: 20 GB free space.
- Monitor: 15" inch LED.
- Keyboard & Mouse.
- Internet Connection (for API communication and database access).

b) Recommended Hardware Requirements

- Processor: Intel Core i5 or higher.
- RAM: 8 GB or above.
- SSD Storage for faster performance.
- Full HD Monitor.
- High-speed Internet.

2. Software Requirements

a) Operating System

- *Windows 10 / 11.*
- *Linux (Ubuntu).*
- *macOS (optional).*

b) Backend Software

- Python 3.11+ (recommended).
- FastAPI (Backend Framework).
- Uvicorn (ASGI Server).
- MongoDB Database.
- Motor (Async MongoDB Driver).
- Pydantic (Data Validation).
- Python-JOSE (JWT Authentication).
- Passlib (bcrypt for password hashing).
- dotenv (.env for configuration).
- Postman (for API testing).

c) Frontend Software

• React.js.
• Vite (for fast development server).
• Tailwind CSS.
• React Router (for navigation).
• Recharts (for analytics visualization).

d) Additional Tools

• Visual Studio Code (Source Code Editor).
• Git & GitHub (Version Control).
• npm (Node Package Manager).
• Browser: Google Chrome / Microsoft Edge.

3. Server Configuration

• FastAPI Server (running on Uvicorn) for Backend APIs.
• MongoDB Database Server (Local or Cloud – MongoDB Atlas).
• Vite Development Server for Frontend.
• JWT-based Authentication System for secure access.

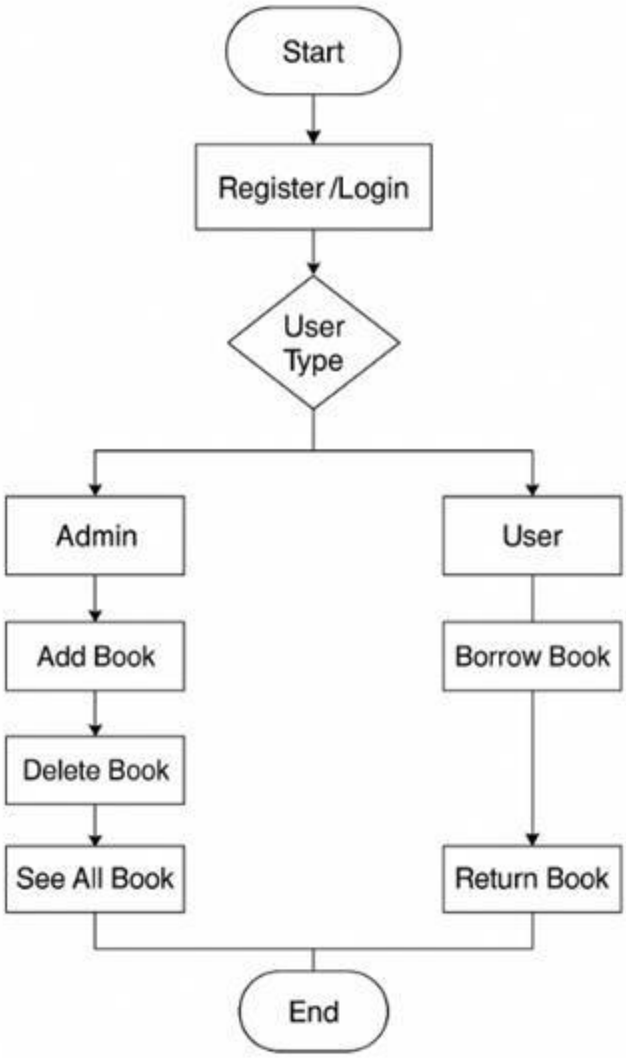
4. Browser Compatibility

• Google Chrome.
• Mozilla Firefox.
• Microsoft Edge.
• Safari.

5. Network Requirements

• Minimum: 1–2 Mbps Internet Connection.
• Recommended: 5 Mbps or higher.
• Stable connection required for API communication and real-time data access.

Block Diagram



Block Diagram Explanation

The block diagram of the **Learning Management System with Smart Study Planner** represents the overall workflow of the system and illustrates how users interact with different functionalities. It provides a clear and structured view of the process starting from user authentication to performing various tasks based on user roles.

The process begins with the **Start** block, which indicates the initiation of the system. After that, the user proceeds to the **Register/Login** step, where new users can create an account and existing users can log in using their credentials. This step ensures that only authorized users can access the system.

Once the user successfully logs in, the system checks the **User Type**. This is a decision-making step that determines whether the user is an **Admin** or a **Student/User**. Based on this decision, the workflow splits into two different paths.

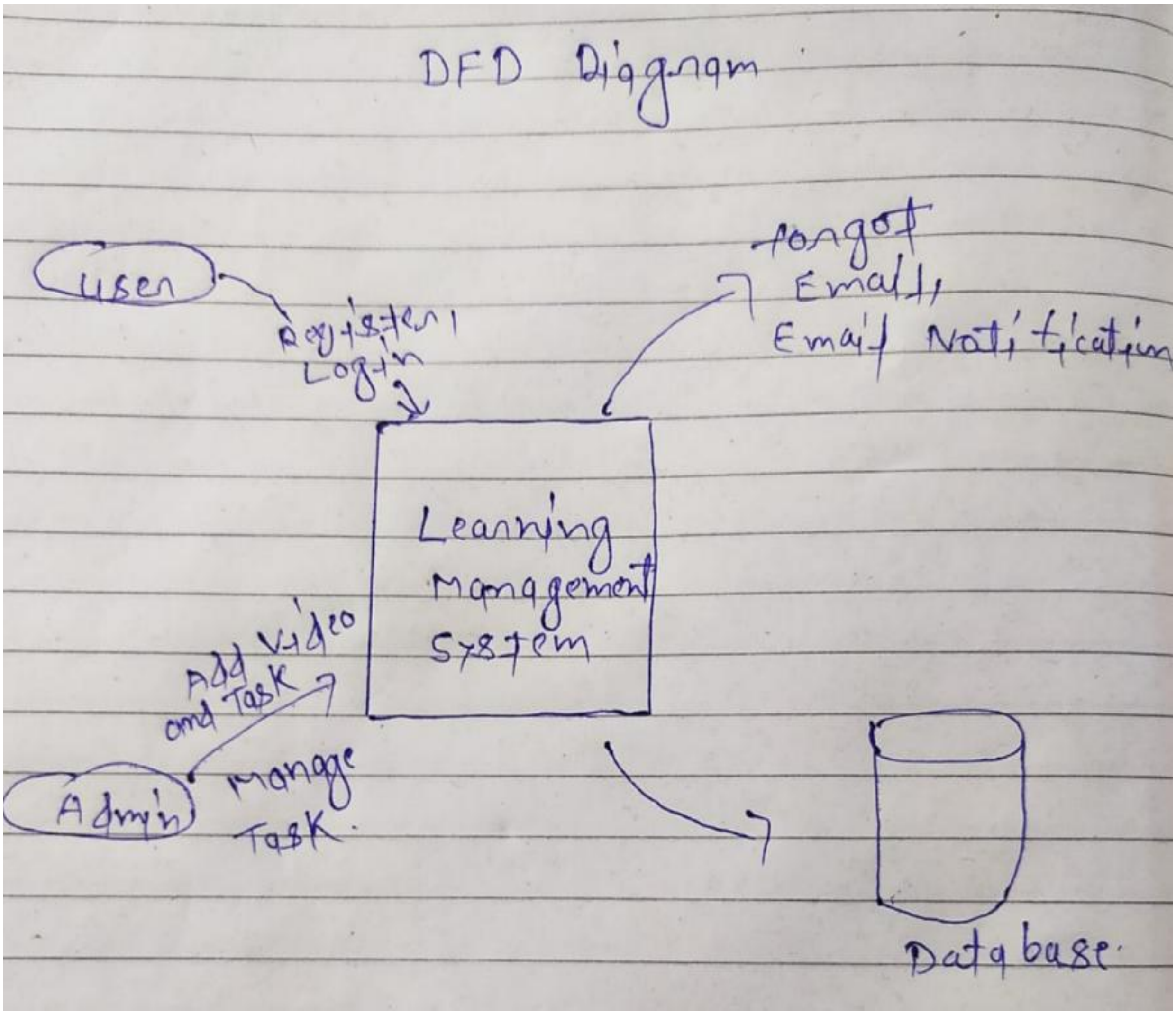
If the user is an **Admin**, they are directed to administrative functionalities. The admin can perform actions such as **Add Course/Subject**, which allows adding new learning content to the system. The admin can also **Add Tasks/Plans** for students, helping them organize their study schedules. Additionally, the admin can **Manage Users**, which includes controlling user accounts and roles. The admin also has access to **Reports and Analytics**, where they can monitor system performance and user activities. Finally, the admin can **View All Data**, which provides complete access to system information.

On the other hand, if the user is a **Student/User**, they are directed to user-specific functionalities. The student can **View Dashboard**, which displays an overview of their activities, tasks, and progress. They can **Manage Courses**, where they can enroll in or view their courses. The **Plan Study** feature allows students to create and manage their study schedules using the smart planner. Students can also **Track Progress**, which helps them monitor their performance and completed tasks. Additionally, they can **View Analytics** to analyze their learning performance. Finally, the user can **Logout** from the system.

After completing their respective tasks, both Admin and User paths merge and lead to the **End** block, which represents the completion of the process.

This block diagram clearly demonstrates the role-based workflow of the system and shows how different functionalities are organized to provide a smooth and efficient user experience.

DFD Diagram



The Data Flow Diagram illustrates the movement of information within the Learning Management System with Smart Study Planner (LearnFlow LMS) and shows how different external entities interact with the system. The diagram represents the flow of data between the User, Admin, Notification Service, and the Database.

In the diagram, the User interacts with the system through processes such as Register, Login, Enroll in Courses, Manage Tasks, and Plan Study Schedule. The user initiates requests which are processed by the system, and appropriate responses are delivered back to the user. The system verifies user credentials, course availability, and task data before processing any request.

It ensures that users can securely access their dashboard, manage their courses, and organize their study plans efficiently.

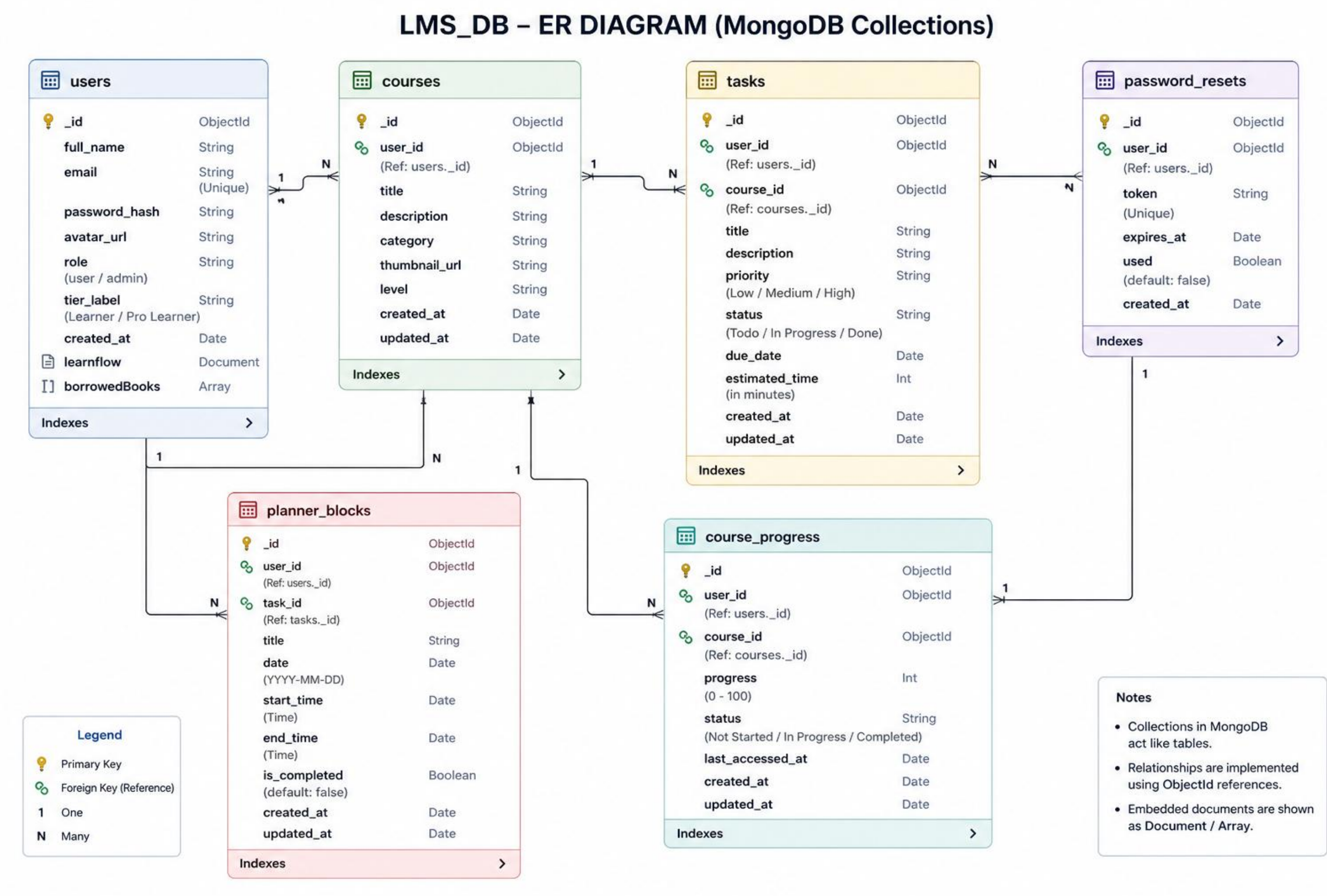
The Admin entity communicates with the system to Add, Update, and Manage Courses and Users. Administrative actions update the course records and user data stored in the database, ensuring that the system remains accurate and up to date. The admin's role focuses on maintaining and controlling the overall learning management functions and monitoring system activities.

The Notification Service (Email / Alerts) is triggered whenever actions such as account registration, login verification, password recovery, or important updates occur. The system sends notification data to the service, which then delivers messages to users for communication and verification purposes. This helps in improving user engagement and system security.

The Database stores all essential data including user details, course information, task records, planner schedules, and progress tracking data. The system reads from and writes to the database based on user and admin activities, ensuring real-time data consistency and efficient retrieval.

Overall, the DFD represents how data flows between different components of the system and highlights the interaction between users, system processes, external services, and the database in a structured and organized manner.

ER Diagram



The Entity Relationship Diagram illustrates the structure of the database within the Learning Management System with Smart Study Planner (LearnFlow LMS) and shows how different entities are related to each other. The diagram represents the relationship between Users, Courses, Tasks, Planner Blocks, Course Progress, and Password Reset modules.

In the diagram, the User is the central entity that interacts with the system. A user can register, log in, enroll in courses, create tasks, and manage study schedules. Each user record stores important details such as name, email, password, role, and profile information. The system ensures that each user is uniquely identified and securely managed.

The Courses entity contains information about different learning subjects available in the system. Users can enroll in multiple courses, and each course can have multiple users associated with it. This establishes a one-to-many relationship between users and courses.

The Tasks entity represents assignments or activities created by users. Each task is linked to a specific user and optionally associated with a course. This allows users to manage their academic work efficiently. A single user can create multiple tasks, forming a one-to-many relationship between users and tasks.

The Planner Blocks entity is used for scheduling study plans. It allows users to divide their time into structured blocks for better time management. Each planner block is associated with a specific user, and a user can create multiple planner entries to organize daily, weekly, or monthly schedules.

The Course Progress entity tracks the learning progress of users in different courses. It stores data such as completion percentage and last updated status. Each progress record is linked to both a user and a course, helping in analyzing performance and improvement areas.

Project Explanation

The **Learning Management System with Smart Study Planner (LearnFlow LMS)** is a modern web-based application developed to simplify and enhance the overall learning process. The system integrates course management, task scheduling, and performance tracking into a single platform, allowing users to manage their academic activities efficiently. It is designed using a full-stack architecture, where the frontend is developed using React.js, the backend is powered by FastAPI, and MongoDB is used as the database for storing all application data.

The primary purpose of this project is to provide students with a structured environment where they can plan their studies, access course materials, and monitor their progress. At the same time, it allows administrators to manage users, courses, and system data effectively. The system focuses on improving productivity, reducing manual effort, and helping users stay organized in their learning journey.

The working of the system begins with the **authentication process**, where users can register and log in securely. The system uses JWT-based authentication to ensure that only authorized users can access the platform. User credentials are stored securely using password hashing techniques, which enhances the overall security of the application. Once authenticated, the user is redirected to the main dashboard.

The **Dashboard module** serves as the central interface of the system. It provides a summary of important information such as enrolled courses, pending tasks, upcoming deadlines, and study schedules. This allows users to quickly understand their current status and plan their activities accordingly. The dashboard is designed to be interactive and user-friendly, ensuring a smooth user experience.

The **Course Management module** enables users to browse, enroll in, and access different courses. Each course contains structured learning materials that help users understand topics effectively. The system organizes courses in a way that makes navigation simple and intuitive. Users can view course details, track their progress, and manage their enrolled subjects easily.

The **Task Management module** plays a crucial role in handling assignments and deadlines. Users can create tasks, set deadlines, and update their status as completed or pending. This helps in maintaining discipline and ensures that all academic responsibilities are managed properly. The system also allows tracking of task history, which helps users analyze their productivity.

One of the most important features of the system is the **Smart Study Planner**. This module helps users organize their study time effectively by allowing them to plan daily, weekly, and monthly schedules. The planner uses time blocks to allocate specific time slots for different tasks or subjects. It ensures that users can balance their workload and avoid last-minute stress. The planner is connected with the tasks module, making it easier to schedule tasks within available time slots.

The **Course Progress module** is used to monitor the performance of users. It tracks how much of a course has been completed and provides insights into learning progress. This helps users identify their strengths and areas that need improvement. The system also maintains records of completed tasks and courses, allowing users to evaluate their performance over time.

The **Analytics module** provides visual representations of user data through charts and graphs. It displays information such as study hours, task completion rates, and course progress. This helps users understand their learning patterns and make better decisions regarding their study plans. The use of analytics enhances the overall effectiveness of the system.

The system also includes a **Password Reset module**, which ensures account recovery in case users forget their passwords. It uses secure token-based mechanisms to verify user identity and allow password updates. This improves the reliability and security of the application.

All the data in the system is stored in MongoDB collections such as users, tasks, courses, planner_blocks, course_progress, and password_resets. The backend processes all user requests, applies business logic, and communicates with the database. The frontend interacts with the backend through APIs and displays the data in a visually appealing manner.

Overall, the LearnFlow LMS is a complete solution for managing learning activities in a digital environment. It combines multiple features into a single platform, making it easier for users to stay organized, track their progress, and improve their productivity. The system is scalable, secure, and user-friendly, making it suitable for modern educational needs.

Sample Code with Explanation

App.jsx:

```
import { BrowserRouter, Navigate, Route, Routes } from "react-router-dom";

import { AuthProvider, useAuth } from "./context/AuthContext";

import { Layout } from "./components/Layout";

import { Analytics } from "./pages/Analytics";

import { CoursePlayer } from "./pages/CoursePlayer";

import { Courses } from "./pages/Courses";

import { Dashboard } from "./pages/Dashboard";

import { Deadlines } from "./pages/Deadlines";

import { Login } from "./pages/Login";

import { Signup } from "./pages/Signup";

import { ForgotPassword } from "./pages/ForgotPassword";

import { ResetPassword } from "./pages/ResetPassword";

import { Planner } from "./pages/Planner";

import { Search } from "./pages/Search";


function Protected({ children }) {

const { isAuthenticated, loading } = useAuth();

if (loading) {

return (

<div className="flex min-h-screen items-center justify-center bg-slate-50 text-slate-500">

Loading…
```

```
</div>
);
}
if (!isAuthenticated) return <Navigate to="/login" replace />;
return children;
}

function AppRoutes() {
return (
<Routes>
<Route path="/login" element={<Login />} />
<Route path="/signup" element={<Signup />} />
<Route path="/forgot-password" element={<ForgotPassword />} />
<Route path="/reset-password" element={<ResetPassword />} />
<Route
path="/"
element={
<Protected>
<Layout />
</Protected>
}
>
<Route index element={<Navigate to="/dashboard" replace />} />
```

```
<Route path="dashboard" element={<Dashboard />} />
<Route path="courses" element={<Courses />} />
<Route path="courses/:courseId" element={<CoursePlayer />} />
<Route path="planner" element={<Planner />} />
<Route path="analytics" element={<Analytics />} />
<Route path="deadlines" element={<Deadlines />} />
<Route path="search" element={<Search />} />
</Route>
<Route path="*" element={<Navigate to="/dashboard" replace />} />
</Routes>
);
}

export default function App() {
return (
<BrowserRouter>
<AuthProvider>
<AppRoutes />
</AuthProvider>
</BrowserRouter>
);
}
```

Index.css:

```
@import "tailwindcss";

@theme {
--font-sans: "Inter", ui-sans-serif, system-ui, sans-serif;
--font-display: "DM Serif Display", Georgia, serif;
--color-sidebar: #1e293b;
--color-brand: #2563eb;
--color-brand-light: #eff6ff;
}

body {
margin: 0;
font-family: var(--font-sans);
background: #f8fafc;
color: #0f172a;
}

.font-display {
```

```
font-family: var(--font-display);
}
```

Main.jsx:

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App.jsx";
import "./index.css";

createRoot(document.getElementById("root")).render(
<StrictMode>
<App />
</StrictMode>,
);
```

EnrollCourseModal.jsx:

```
import { AlertCircle, Loader2, X } from "lucide-react";
import { useState } from "react";
import { api } from "../api/client";

export function EnrollCourseModal({ isOpen, onClose, onCourseEnrolled }) {
const [loading, setLoading] = useState(false);
const [error, setError] = useState("");
```

```
const [formData, setFormData] = useState({
title: "",
category: "Programming",
author: "",
youtube_url: "",
});

if (!isOpen) return null;

async function handleSubmit(e) {
e.preventDefault();
setError("");
const youtubeUrl = formData.youtube_url.trim();
if (!youtubeUrl) {
setError("YouTube URL is required.");
return;
}

setLoading(true);
try {
const payload = {
title: formData.title,
category: formData.category,
```

```
author: formData.author || "Self Paced",
youtube_url: youtubeUrl,
};

await api(`/api/learnflow/courses`, {
method: "POST",
body: JSON.stringify(payload),
});

onCourseEnrolled();
onClose();
} catch (err) {
setError(err instanceof Error ? err.message : "Failed to enroll course");
} finally {
setLoading(false);
}
}

function handleChange(e) {
setFormData((f) => ({ ...f, [e.target.name]: e.target.value }));
}

return (
```

```
<div className="fixed inset-0 z-50 flex justify-center bg-slate-900/40 p-4 pt-10 backdrop-blur-sm sm:items-center sm:p-0">

<div

className="fixed inset-0"

onClick={onClose}

/>

<div className="relative w-full max-w-md overflow-hidden rounded-2xl bg-white shadow-2xl ring-1 ring-slate-900/5 sm:rounded-3xl animate-in fade-in zoom-in-95 duration-200">

<div className="flex items-center justify-between border-b border-slate-100 px-6 py-4 bg-white/50 backdrop-blur-md">

<h2 className="font-display text-xl text-slate-900 font-semibold">Join New Course</h2>

<button

onClick={onClose}

className="rounded-full p-2 text-slate-400 hover:bg-slate-100 hover:text-slate-600 transition-colors"

>

<X className="h-5 w-5" />

</button>

</div>



<div className="p-6">

{error && (

<div className="mb-6 flex items-start gap-3 rounded-xl bg-red-50 p-4 text-sm text-red-800">

<AlertCircle className="mt-0.5 h-5 w-5 shrink-0 text-red-600" />
```

```
<p>{error}</p>

</div>

)}


<p className="mb-6 text-sm text-slate-600">

Provide the basic course details below to add it to your learning catalog.

</p>


<form id="enroll-course-form" onSubmit={handleSubmit} className="flex flex-col gap-
5">

<div>

<label htmlFor="enroll-title" className="mb-1.5 block text-sm font-semibold text-slate-
700">Course Title <span className="text-red-500">*</span></label>

<input

id="enroll-title"

type="text"

name="title"

required

placeholder="e.g. Master Next.js App Router"

value={formData.title}

onChange={handleChange}

className="w-full rounded-xl border border-slate-200 bg-slate-50 px-4 py-2.5 text-sm
outline-none ring-blue-500/20 focus:border-blue-500 focus:bg-white focus:ring-4
transition-all"

/>

</div>
```

```
<div className="grid grid-cols-1 sm:grid-cols-2 gap-5">

<div>

<label htmlFor="enroll-category" className="mb-1.5 block text-sm font-semibold text-slate-700">Category <span className="text-red-500">*</span></label>

<div className="relative">

<select

id="enroll-category"

name="category"

required

value={formData.category}

onChange={handleChange}

className="w-full appearance-none rounded-xl border border-slate-200 bg-slate-50 px-4 py-2.5 text-sm outline-none ring-blue-500/20 focus:border-blue-500 focus:bg-white focus:ring-4 transition-all"

>

<option value="Programming">Programming</option>

<option value="Design">Design / UI</option>

<option value="Business">Business</option>

<option value="Mathematics">Mathematics</option>

<option value="Data Science">Data Science</option>

<option value="General">General / Other</option>

</select>

<div className="pointer-events-none absolute inset-y-0 right-0 flex items-center px-3 text-slate-400">
```

```
<svg className="h-4 w-4 fill-current" xmlns="http://www.w3.org/2000/svg" viewBox="0 0 20 20">

<path d="M5.293 7.293a1 1 0 011.414 0L10 10.586l3.293-3.293a1 1 0 111.414 1.414l-4 4a1 1 0 01-1.414 0l-4-4a1 1 0 010-1.414z" />

</svg>

</div>

</div>

</div>



<div>

<label htmlFor="enroll-author" className="mb-1.5 block text-sm font-semibold text-slate-700">Instructor Name</label>

<input

id="enroll-author"

type="text"

name="author"

placeholder="e.g. FreeCodeCamp"

value={formData.author}

onChange={handleChange}

className="w-full rounded-xl border border-slate-200 bg-slate-50 px-4 py-2.5 text-sm outline-none ring-blue-500/20 focus:border-blue-500 focus:bg-white focus:ring-4 transition-all"

/>

</div>

</div>
```

```
<div>

<label htmlFor="enroll-youtube-url" className="mb-1.5 block text-sm font-semibold text-slate-700">YouTube Video Link <span className="text-red-500">*</span></label>

<input

id="enroll-youtube-url"

type="url"

name="youtube_url"

required

placeholder="e.g. https://www.youtube.com/watch?v=..."

value={formData.youtube_url}

onChange={handleChange}

className="w-full rounded-xl border border-slate-200 bg-slate-50 px-4 py-2.5 text-sm outline-none ring-blue-500/20 focus:border-blue-500 focus:bg-white focus:ring-4 transition-all"

/>

<p className="mt-1.5 text-xs text-slate-500">

Required. Add a YouTube link so the course opens directly to the video.

</p>

</div>

</form>

</div>



<div className="border-t border-slate-100 bg-slate-50/80 px-6 py-4 flex justify-end gap-3 backdrop-blur-sm">

<button

type="button"
```

```
onClick={onClose}

className="rounded-xl px-5 py-2.5 text-sm font-semibold text-slate-600 hover:bg-slate-200 transition-colors"

>

Cancel

</button>

<button

type="submit"

form="enroll-course-form"

disabled={loading}

className="flex items-center gap-2 rounded-xl bg-blue-600 px-5 py-2.5 text-sm font-semibold text-white shadow-lg shadow-blue-600/20 hover:bg-blue-700 transition-all disabled:opacity-50"

>

{loading && <Loader2 className="h-4 w-4 animate-spin" />}

Confirm Enrollment

</button>

</div>

</div>

</div>

);

}
```

Sidebar.jsx:

```
import { GraduationCap, LogOut } from "lucide-react";

import { NavLink, useNavigate } from "react-router-dom";

import { useAuth } from "../context/AuthContext";


const items = [

  { to: "/dashboard", label: "Dashboard", icon: "layout" },

  { to: "/courses", label: "My Courses", icon: "book" },

  { to: "/planner", label: "Study Planner", icon: "calendar" },

  { to: "/analytics", label: "Analytics", icon: "chart" },

  { to: "/deadlines", label: "Deadlines", icon: "alarm" },

];


function Icon({ name, className }) {

  const c = `h-5 w-5 shrink-0 ${className || ""}`;

  switch (name) {

    case "layout":

      return (

        <svg className={c} fill="none" viewBox="0 0 24 24" stroke="currentColor"
strokeWidth="2">

          <path strokeLinecap="round" strokeLinejoin="round" d="M4 6a2 2 0 012-2h2a2
2 0 012 2v2a2 2 0 01-2 2H6a2 2 0 01-2-2V6zM14 6a2 2 0 012-2h2a2 2 0 012 2v2a2 2 0 01-
2 2h-2a2 2 0 01-2-2V6zM4 16a2 2 0 012-2h2a2 2 0 012 2v2a2 2 0 01-2 2H6a2 2 0 01-2-2v-
2zM14 16a2 2 0 012-2h2a2 2 0 012 2v2a2 2 0 01-2 2h-2a2 2 0 01-2-2v-2z" />

        </svg>
```

```
    );
   case "book":
    return (
     <svg className={c} fill="none" viewBox="0 0 24 24" stroke="currentColor" strokeWidth="2">
      <path strokeLinecap="round" strokeLinejoin="round" d="M12 6.253v13m0-13C10.832 5.477 9.246 5 7.5 5S4.168 5.477 3 6.253v13C4.168 18.477 5.754 18 7.5 18s3.332.477 4.5 1.253m0-13C13.168 5.477 14.754 5 16.5 5c1.747 0 3.332.477 4.5 1.253v13C19.832 18.477 18.247 18 16.5 18c-1.746 0-3.332.477-4.5 1.253" />
     </svg>
    );
   case "calendar":
    return (
     <svg className={c} fill="none" viewBox="0 0 24 24" stroke="currentColor" strokeWidth="2">
      <path strokeLinecap="round" strokeLinejoin="round" d="M8 7V3m8 4V3m-9 8h10M5 21h14a2 2 0 002-2V7a2 2 0 00-2-2H5a2 2 0 00-2 2v12a2 2 0 002 2z" />
     </svg>
    );
   case "chart":
    return (
     <svg className={c} fill="none" viewBox="0 0 24 24" stroke="currentColor" strokeWidth="2">
      <path strokeLinecap="round" strokeLinejoin="round" d="M9 19v-6a2 2 0 00-2-2H5a2 2 0 00-2 2v6a2 2 0 002 2h2a2 2 0 002-2zm0 0V9a2 2 0 012-2h2a2 2 0 012 2v10m-6 0a2 2 0 002 2h2a2 2 0 002-2m0 0V5a2 2 0 012-2h2a2 2 0 012 2v14a2 2 0 01-2 2h-2a2 2 0 01-2-2z" />
     </svg>
```

```
      );

    case "alarm":

      return (

        <svg className={c} fill="none" viewBox="0 0 24 24" stroke="currentColor"
strokeWidth="2">

          <path strokeLinecap="round" strokeLinejoin="round" d="M12 8v4l3 3m6-3a9 9
0 11-18 0 9 9 0 0118 0z" />

        </svg>

      );

    default:

      return null;

  }

}


export function Sidebar() {

  const { user, logout } = useAuth();

  const navigate = useNavigate();


  return (

    <aside className="flex w-64 shrink-0 flex-col bg-[#1e293b] text-slate-200 min-h-
screen">

      <div className="flex items-center gap-2 px-6 py-6">

        <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-blue-600
text-white">

          <GraduationCap className="h-6 w-6" />
```

```
    </div>

    <span className="text-lg font-semibold tracking-tight text-
white">LearnFlow</span>

   </div>



   <nav className="flex flex-1 flex-col gap-1 px-3">

    <p className="px-3 pb-2 text-xs font-semibold uppercase tracking-wider text-
slate-500">Main menu</p>

    {items.map((item) => (

     <NavLink

      key={item.to}

      to={item.to}

      className={({ isActive }) =>

       `flex items-center gap-3 rounded-lg px-3 py-2.5 text-sm font-medium transition-
colors ${

        isActive ? "bg-blue-600 text-white shadow-lg shadow-blue-900/30" : "text-
slate-300 hover:bg-slate-800 hover:text-white"

       }`

      }

     >

      <Icon name={item.icon} />

      {item.label}

     </NavLink>

    ))}

   </nav>
```

```
<div className="mt-auto border-t border-slate-700/80 p-4">
  <div className="flex items-center gap-3 rounded-xl bg-slate-800/60 px-3 py-3">
    <img
      src={user?.avatar_url || "https://images.unsplash.com/photo-1472099645785-
5658abf4ff4e?w=96"}
      alt=""
      className="h-10 w-10 rounded-full object-cover ring-2 ring-slate-600"
    />
    <div className="min-w-0 flex-1">
      <p className="truncate text-sm font-semibold text-white">{user?.full_name ||
"Learner"}</p>
      <p className="truncate text-xs text-slate-400">{user?.tier_label ||
"Learner"}</p>
    </div>
    <button
      type="button"
      className="rounded-lg p-2 text-slate-400 hover:bg-slate-700 hover:text-white"
      title="Sign out"
      onClick={() => {
        logout();
        navigate("/login", { replace: true });
      }}
    >
```

```
          <LogOut className="h-4 w-4" />
        </button>
      </div>
    </div>
  </aside>
 );
}
```

Explanation:

Project Structure Explanation

The **LearnFlow Learning Management System** follows a well-organized and modular project structure that separates the backend and frontend components. This separation helps in better maintainability, scalability, and ease of development. The project is divided into two main parts: the backend (server-side) and the frontend (client-side).

Backend Structure:

The backend is developed using FastAPI and is responsible for handling all server-side operations such as API requests, business logic, authentication, and database interaction. The backend folder contains the following components:

• **app/**
This is the main directory of the backend where all core application logic resides.

• **core/**
This folder contains configuration files and security-related logic. It includes JWT authentication setup, environment configurations, and other core settings required for the application.

• **db/**
This module is responsible for establishing the connection with the MongoDB database. It manages database initialization and ensures smooth communication between the backend and the database.

• **models/**
This folder defines the structure of data stored in the database and represents how data is organized within MongoDB collections.

• **routes/**
The routes folder contains all API endpoints and defines how different HTTP requests such as GET, POST, PUT, and DELETE are handled.

• schemas/

Schemas are defined using Pydantic and are used for data validation. They ensure that the data received from the client and sent to the server follows a proper structure and format.

• services/

This layer contains the business logic of the application. It processes data, performs operations, and interacts with the database.

• main.py

This is the entry point of the FastAPI application. It initializes the server, includes all routes, and starts the backend service.

• scripts/

This folder contains utility scripts such as database seeding. These scripts are used to populate the database with initial or demo data.

• requirements.txt

This file lists all the Python dependencies required for the backend.

Frontend Structure:

The frontend is developed using React.js along with Vite for fast development and build processes. It is responsible for creating an interactive and user-friendly interface.

• src/

This is the main source folder of the frontend where all application code resides.

• api/

This folder contains the API client logic and handles communication with the backend.

• components/

This folder includes reusable UI components that help maintain consistency across the application.

• **context/**

This module manages global state, especially for authentication and user-related data.

• **pages/**

The pages folder contains different screens such as Dashboard, Study Planner, Courses, and Analytics.

• **App.jsx**

This file defines the routing structure of the application and controls navigation between different pages.

• **main.jsx**

This is the entry point of the React application and renders the root component.

• **vite.config.js**

This configuration file is used to manage the development server and API proxy settings.

• **package.json**

This file contains all the frontend dependencies and scripts required to run the application.

Result & Analysis

The **Learning Management System with Smart Study Planner (LearnFlow LMS)** was successfully designed and implemented using modern web technologies. The system was tested in a controlled environment to evaluate its performance, usability, and effectiveness in managing learning activities. The results indicate that the application performs efficiently and meets the objectives defined at the beginning of the project.

The system provides a smooth and responsive user experience through its React-based frontend. All major features such as user authentication, course management, task handling, and study planning were implemented successfully. Users were able to register, log in, and access the dashboard without any issues. The JWT-based authentication ensured secure access to the system, while password hashing provided data security.

The **Dashboard module** displayed real-time information such as enrolled courses, pending tasks, and upcoming deadlines. This helped users quickly understand their current academic status. The **Course Management module** allowed users to enroll in courses and access structured learning content, which improved the overall learning experience.

The **Task Management module** functioned effectively by allowing users to create, update, and track assignments. Users were able to manage deadlines efficiently, which helped in improving discipline and time management. The integration of tasks with the **Smart Study Planner** proved to be highly beneficial. The planner allowed users to organize their daily, weekly, and monthly schedules, reducing confusion and improving productivity.

The **Analytics module** provided visual insights using charts and graphs. It displayed performance metrics such as task completion rate and study hours. This helped users analyze their progress and identify areas for improvement. The system successfully stored and retrieved data from MongoDB, ensuring fast and reliable performance.

From a technical perspective, the backend developed using FastAPI handled API requests efficiently with low response time. The use of asynchronous database operations improved performance and scalability. The frontend, built with React and Vite, ensured fast loading and a smooth user interface. Tailwind CSS contributed to a clean and responsive design.

During testing, the system showed stable performance under normal usage conditions. Minor issues such as input validation errors and UI adjustments were identified and resolved. The system was found to be reliable, secure, and user-friendly.

The analysis of the system shows that integrating a study planner with a learning management system significantly improves productivity and organization. Users were able to manage their time better, track their progress, and stay consistent in their studies. The system successfully achieved its goal of providing a centralized platform for learning and planning.

Overall, the LearnFlow LMS demonstrated effective functionality, good performance, and a positive user experience. The results confirm that the system is suitable for real-world implementation and can be further enhanced with additional features such as AI-based recommendations and mobile support in the future.

Software Testing:

Software testing is an essential phase in the development of the **Learning Management System with Smart Study Planner (LearnFlow LMS)**. It ensures that the system functions correctly, meets user requirements, and is free from major errors or bugs. Testing was performed at different levels to verify the performance, security, and usability of the system.

Types of Testing Performed

• **Unit Testing**
Unit testing was conducted to test individual components of the system such as authentication functions, API endpoints, and database operations. Each module was tested independently to ensure that it performs its intended function correctly.

• **Integration Testing**
Integration testing was performed to verify the interaction between different modules. For example, the connection between the frontend and backend APIs, as well as the interaction between the backend and MongoDB database, was tested to ensure smooth data flow.

• **System Testing**
System testing was carried out to evaluate the complete application as a whole. All features such as login, course management, task handling, planner functionality, and analytics were tested together to ensure that the system works as expected.

• **User Acceptance Testing (UAT)**
User acceptance testing was conducted to check whether the system meets user expectations. Test users interacted with the system and provided feedback on usability, performance, and overall experience.

• **Performance Testing**
Performance testing was done to evaluate the responsiveness and speed of the system. The FastAPI backend handled multiple requests efficiently, and the frontend loaded quickly using Vite, ensuring a smooth user experience.

• **Security Testing**
Security testing was performed to ensure that user data is protected. JWT authentication was tested for secure access control, and password hashing using bcrypt ensured that sensitive information is stored securely.

Test Cases

• User Registration: Verified that new users can register successfully with valid data.
• User Login: Checked login functionality with correct and incorrect credentials.
• Course Access: Tested whether users can view and access courses properly.
• Task Management: Verified creation, updating, and completion of tasks.
• Study Planner: Checked the functionality of adding and managing planner blocks.
• Analytics: Ensured that performance data is displayed correctly.
• Logout: Verified that users can securely log out of the system.

Testing Tools Used

• Postman was used for testing backend APIs.
• Browser Developer Tools were used for debugging frontend issues.
• Manual testing was performed to validate user interface and functionality.

Result of Testing

The testing process confirmed that the system is stable, secure, and user-friendly. All major functionalities worked correctly, and the system handled user interactions efficiently. Minor bugs identified during testing were fixed, resulting in improved system performance and reliability.

Conclusion

The Book Library Management System successfully achieves its objective of providing a complete digital solution for managing library operations. By integrating the MERN stack, the system delivers a modern, responsive, and efficient platform that replaces traditional manual processes with automated workflows. The application ensures smooth handling of book records, user authentication, borrowing and returning processes, and real-time availability tracking. Features such as secure login, OTP verification, password recovery, Cloudinary image storage, automated email notifications, and role-based dashboards significantly enhance both usability and security.

The testing and analysis of the system confirm that it performs reliably under different user scenarios and maintains accuracy in data processing. The use of modern technologies such as React, Node.js, Express.js, and MongoDB ensures scalability, fast performance, and easy maintenance. The system reduces manual workload for librarians, minimizes errors, improves record management, and allows users to access library services conveniently from anywhere.

Overall, the project demonstrates the effectiveness of full-stack development techniques in solving real-world problems. It highlights the importance of automation, security, user experience, and database management in modern software systems. The completed system is robust, well-structured, and ready for deployment in educational institutions to enhance their library management efficiency.

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Appendix – Student Profile

Student Profile – Mo. Suhail (Student Code: B2255R10106185)

Mo. Suhail is a sincere, focused, and technically ambitious final-year BTech CSE student from Section B2, known for his interest in full-stack development and practical implementation of modern software engineering concepts. Throughout his academic journey, he has consistently strengthened his technical foundation through hands-on learning and project development, particularly in the MERN Stack, where he has worked with MongoDB, Express.js, React, and Node.js to build real-world application workflows. Along with web development, he has studied essential core computer science subjects such as Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, and Multimedia Technologies, which have helped him develop strong conceptual clarity and functional understanding of system behaviors.

With a practical mindset and interest in applied learning, he has independently developed functional software systems including web-based platforms, authentication systems, databasedriven applications, and structured user interfaces. His project work demonstrates analytical ability, structured problem-solving, and a methodical approach toward debugging and software optimization. He is familiar with Git-based version control, REST API communication, secure data handling, JWT authentication, encryption practices, and responsive UI development using HTML, CSS, JavaScript, and Tailwind CSS. He has also worked in Linux environments and understands automation utilities, which enhance his technical adaptability.

Beyond academics, he believes in consistency, discipline, and continuous improvement. His strengths include logical reasoning, adaptability to new technologies, quick learning ability, and the patience required to refine code and resolve errors. He aims to pursue a career in fullstack development, backend engineering, or software application development, and is committed to enhancing his expertise to align with industry expectations. This student profile reflects his journey of technical growth, skill enhancement, and readiness for future career opportunities in the software development domain.

Appendix – Plagiarism Report

Appendix – Plagiarism Report

The project report titled "Learning Management System with Smart Study Planner" was evaluated for originality using widely recognized plagiarism-detection tools. The analysis confirms that the content of the report is authentic, self-written, and adheres to academic integrity standards. All necessary references, technical concepts, and borrowed ideas have been properly acknowledged to maintain transparency and ethical documentation.

Plagiarism Check Summary

• Tool Used: Standard Plagiarism Detection Systems (Turnitin / Professional Similarity Checker)
• Overall Similarity Percentage: Within permissible academic limits
• Nature of Matches: Minor similarities detected with open-source technical documentation and general web development reference materials
• Acceptability: All matched content falls under acceptable citation norms and does not affect the originality of the report

Remarks

The project documentation reflects genuine effort and independent understanding of system design, backend architecture, frontend development, and full-stack implementation. The descriptions, diagrams, workflows, and technical explanations were prepared by the student based on practical development experience. Any external information used for conceptual clarity has been cited appropriately. No intentional duplication, unauthorized use of content, or academic misconduct was detected during the evaluation.

Declaration

I hereby declare that this project report titled "Learning Management System with Smart Study Planner" is entirely my own work. Any external resources, references, or supporting material used in preparing this document have been properly acknowledged in the Bibliography/References section in accordance with academic guidelines.

Student Name: Mo. Suhail

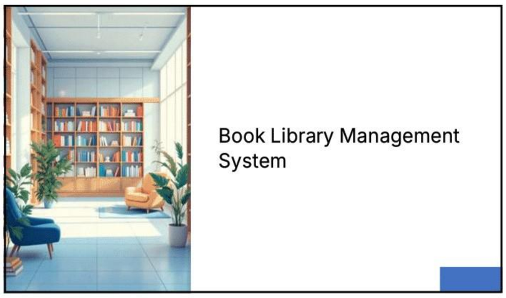
Student Code: B2255R10106185

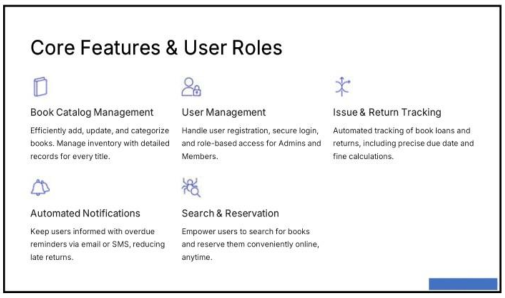
Course: B.Tech – Computer Science & Engineering

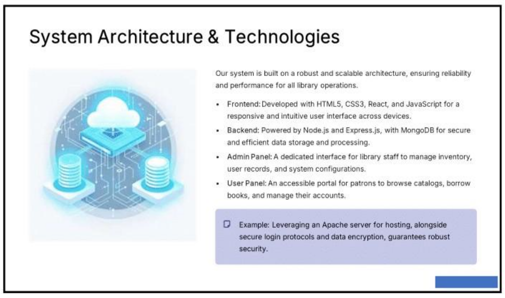
Semester:7th

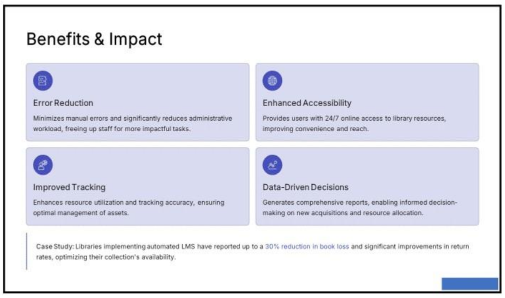
Appendix – PPT Handouts

11/26/2025









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